§6 Minimal structure of ACS relations — part 2 of 4

Reference: subsec:rel_bidirect, subsec:ro_quality up to the formalization of Q

Source: paper/section6_relational.tex, lines 712–1331 (bidirectional composition and its auxiliary notions; the opening of relation quality; the evaluation rule, the evaluable domain and the definition of Q with its remarks), at commit e6e12cd. Status: draft. Part 1 is 06_relational_1.md; part 3 (06_relational_3.md, 1332–1791: sign classification of Q through the relation to T-STAB) and part 4 (06_relational_4.md, 1792–2171) follow.

Bidirectional composition

Reference: subsec:rel_bidirect

A bidirectional composition is an ordered pair of two unidirectional records for objects that can both be treated as ACSs, formed at times where both directions are well-formed. It is not an operation that constructs a new ACS or a common state space by combining the internal states of the two individuals.

Assumption (ass:bidirect_scope): standard objects of bidirectional composition. The standard objects are pairs where $Y_j \in \mathcal{Y}_{\mathrm{ACS},-i}$ is treated as an ACS individual Σ_j and the reverse description $\Sigma_i \in \mathcal{Y}_{\mathrm{ACS},-j}$ is also admitted. Composition is made only when the six-component descriptors of both directions are each defined at a common positive window length L and the same time t. The condition does not require that the two individuals adopt the same local size indicator $\mathrm{sz}_i = \mathrm{sz}_j$ or the same category, cost or perturbation values. The existence of a descriptor in one direction does not require the existence of one in the reverse direction. Object identification and relation continuity on both sides are treated as records of each individual. Even if both are confirmed to refer to the same object, it is not assumed that their internal models hold the same content. Introduction status: S (structural, restriction of the object domain). Strength: weak. Relied on in subsec:rel_bidirect and subsec:rel_categories.

Definition (def:bidirect_relation): bidirectional composite relation (classification C). Treat $Y_j \in \mathcal{Y}_{\mathrm{ACS},-i}$ as an ACS individual Σ_j, with $\Sigma_i \in \mathcal{Y}_{\mathrm{ACS},-j}$ in the reverse direction. After confirming the evaluation domain of the local size indicator and the domain of ordinary ACS state quantities of each individual, if both $\mathrm{Rel}_{i\leftarrow j,t}$ and $\mathrm{Rel}_{j\leftarrow i,t}$ are well-formed at a common positive window length L and time t, the bidirectional composite relation is the ordered pair $\mathrm{Rel}_{ij,t} = (\mathrm{Rel}_{i\leftarrow j,t}, \mathrm{Rel}_{j\leftarrow i,t})$. Both directions use the same time range $W_{L,t}$, but tracking of the same object, relation continuity and the domain of ordinary ACS state quantities of each individual must be satisfied separately. When the trajectory records of §7 are adopted for both, with σ_i and σ_j the records of Σ_i and Σ_j, $W_{L,t} \subseteq I_{\mathrm{ACS}}(\sigma_i)$ and $W_{L,t} \subseteq I_{\mathrm{ACS}}(\sigma_j)$ are required respectively; this is not a claim that one trajectory serves as the ordinary state of both individuals. Even if the whole window lies in both valid time domains, the acquisition conditions for the three differences and the first three components are separately needed. The definition does not newly construct the internal dynamics of an external object and only pairs the descriptions of two ACS sides. Since the first and second components contain observed changes on different individuals' sides, the definition does not include as a requirement an increase of feasibility-related quantities of both, mutual benefit, or the establishment of an organization-level integrated structure. ij is a marker distinguishing the pair of objects and does not denote a product or a numerical operation.

Asymmetry of the bidirectional composite relation (rem:bidirect_asymmetry). $\mathrm{Rel}_{ij,t}$ is an ordered pair of two unidirectional descriptors, and $\mathrm{Rel}_{i\leftarrow j,t} = \mathrm{Rel}_{j\leftarrow i,t}$ is not assumed. Even with the same category, cost, perturbation and the three differences can differ as values on different individuals' sides. That both directions are recorded does not mean the same action or mutual benefit exists on both. Since the three differences of each direction are computed from time series belonging to each individual, the two numerical differences are not added unconditionally or read as a feasibility indicator common to both. Even if one descriptor is defined, if the record needed for the reverse direction is missing, no standard bidirectional composition is placed. For example, even when the same relation category is recorded in both

directions, one side's resource load being larger or the differences of the effective time horizon having opposite signs is not formally excluded. Bidirectionality is not read as an assumption of symmetry.

Definition (`def:relation_direction`): direction indicator (classification C). When a bidirectional composite relation is defined, an auxiliary indicator of the principal direction of description for interpretation is $\mathrm{dir}_{ij,t} \in \{i \leftarrow j,\ i \rightarrow j,\ i \leftrightarrow j,\ \text{undetermined}\}$, meaning respectively that the description from j to i is principal, the reverse is principal, both are principal, and the principal direction is not judged by the rules of this section. Because the criterion, threshold and weighting of "principal" are not fixed, it is not claimed that the indicator is uniquely computed from the two unidirectional descriptors. When no criterion is adopted, undetermined may be recorded; it is neither a numerical judgment that both directions are zero nor a classification that the two have no relation. The value of the direction indicator is not included in the six-component descriptors. What counts as principal can depend on the object of record and the comparison standard adopted. Showing the range of values does not give a unique computing rule for the principal direction.

Auxiliary status of the direction indicator (`rem:dir_auxiliary`). The direction indicator is a descriptive aid; it does not replace the two unidirectional descriptors and does not prescribe blockage, collapse or a direction of improvement of feasibility.

Undefined part of direction judgment (`rem:dir_undefined`). A quantitative rule for the principal direction (which of cost, perturbation and the three differences to weigh and how much) is delegated to Phase 1.α (`subsec:phase1a`). undetermined means neither absence of a relation, symmetry of the two directions, nor numerical zero. $\mathrm{dir}_{ij,t}$ is not a quantity identifying the direction of causation between the objects.

Definition (`def:homo_hetero_relation`): homogeneous and heterogeneous mutual relations. For $\mathrm{cat}_{i\leftarrow j,t}, \mathrm{cat}_{j\leftarrow i,t} \in \mathcal{C}_{\rightarrow}$ contained in the two unidirectional descriptors of $\mathrm{Rel}_{ij,t}$, the case $\mathrm{cat}_{i\leftarrow j,t} = \mathrm{cat}_{j\leftarrow i,t}$ is called a homogeneous mutual relation and $\mathrm{cat}_{i\leftarrow j,t} \neq \mathrm{cat}_{j\leftarrow i,t}$ a heterogeneous mutual relation. This distinction only examines agreement or disagreement of labels when the category of each direction is already recorded, and is not the same operation as judging the bidirectional-establishment categories.

Limits of the homogeneous/heterogeneous classification (`rem:homo_hetero_rel`). Homogeneity means agreement of labels only, not agreement of cost, intensity, duration or the three differences. The systematization of response-side categories (being dominated, being managed and the like) is delegated to Phase 1.α. Even in the homogeneous case, the costs, perturbations and the three differences of both sides can differ; in the heterogeneous case the possibility that both differences show the same sign is not excluded. To use additional labels such as "being manipulated", "being exploited" or "being used", establishment conditions of the response side must be defined separately.

Categories without a homogeneous mutual form (`rem:no_mutual_form`). Even when domination or parasitism is recorded in both directions, it is not immediately defined as an independent established category of "mutual domination" or "mutual parasitism". The composite categories and establishment rules needed are objects of Phase 1.α. Organizing into descriptions such as competitive control, mutual constraint or mutually eroding dependence is conceivable, but this paper does not define them as formal categories beyond the 18. The form that the same unidirectional label appears twice is distinguished from the establishment conditions specific to the natural-language name "mutual".

Definition (`def:bidirectional_category`): bidirectional establishment categories (classification C). When $\mathrm{Rel}_{ij,t}$ is defined and a label is attached to its bidirectional mode of establishment, the range is $\mathrm{cat}_{\leftrightarrow,ij,t} \in \mathcal{C}_{\leftrightarrow} = \{\mathrm{coop}, \mathrm{sym}, \mathrm{conf}\}$, whose formal definition is in `def:category_set`. These are bidirectional labels for cooperation, symbiosis and conflict and are distinguished from the category component $\mathrm{cat}_{i\leftarrow j,t} \in \mathcal{C}_{\rightarrow}$ of a unidirectional descriptor. The section gives neither establishment conditions for the bidirectional labels nor derivation rules from two unidirectional labels, so whether to adopt this auxiliary label is a descriptive choice separate from the well-formedness of the bidirectional composition. Even if the homogeneous/heterogeneous classification is fixed, none of the three is decided by it, and $\mathrm{cat}_{\leftrightarrow,ij,t}$ is not assigned to every pair for which $\mathrm{Rel}_{ij,t}$ is defined. "Undetermined" is not added as a new 19th relation category.

Bidirectionality of cooperation, symbiosis and conflict (`rem:cooperation_bidirectional`). The three labels are not included in the set of unidirectional categories and are placed at the level of comparing the records of two directions. The existence of a bidirectional composite relation alone does not determine which of the three holds. Giving establishment conditions and thresholds for contribution, mutuality and opposition is a task of Phase 1.α. For example, that support is recorded in one direction does not show support in the reverse direction; even when support is recorded in both directions, that alone is not a sufficient condition for cooperation. What mutual actions are called cooperation, symbiosis or conflict can be judged only once the rule comparing the bidirectional descriptions is made concrete.

Special feature concerning the direction of competition (`rem:competition_directionality`). Competition comp is a unidirectional category and can record the competitive action of one side only. If $\mathrm{cat}_{i\leftarrow j,t} = \mathrm{cat}_{j\leftarrow i,t} = \mathrm{comp}$ it can be recorded as a homogeneous mutual relation but is not made a separate bidirectional establishment category. When one object records a competitive action and none of the same kind is recorded on the other side, there is no need to move this unidirectional label to a bidirectional one. The quantitative distinction of intensity and establishment threshold when competition appears on both sides is delegated to Phase 1.α.

Bidirectional establishment categories versus homogeneous mutual relations (`rem:bidirect_vs_homo`). The homogeneous/heterogeneous distinction (`def:homo_hetero_relation`) stating agreement of two unidirectional labels and the bidirectional establishment labels (`def:bidirectional_category`) are different axes of description. The section has no rule deriving establishment of cooperation merely from two homogeneous support labels. Conversely, attaching a bidirectional label does not make the two unidirectional records unnecessary. The section does not define a total map from $\mathcal{C}_{\rightarrow} \times \mathcal{C}_{\rightarrow}$ to $\mathcal{C}_{\leftrightarrow}$; whether two unidirectional labels are homogeneous or heterogeneous is not a sufficient condition for deciding which bidirectional label holds. Even if a bidirectional label is recorded, the pair of unidirectional categories cannot be uniquely recovered from it alone.

Limits for bidirectional relations with non-ACS objects (`rem:nACS_limit`). For a non-ACS agent or a non-agent object, the section defines no state space, feasible action set or effective freedom on the side of Σ_j; $\mathrm{Rel}_{j\leftarrow i,t}$ is therefore not generated in the ACS-type six-component form. The unidirectional description on the side of Σ_i remains allowed. A bidirectional form that makes explicit the internal structure of the non-ACS side is outside this paper. Even if a non-ACS agent has some agency, this paper does not assign to it the 7-tuple or the three mechanisms of an ACS. For interaction with objects, records on the individual side can be made, but a reverse six-component record is not made by assuming a feasible action set on the object side.

Bidirectional composition and the blockage and collapse section (`rem:bidirect_to_sec7`). Either of the two unidirectional descriptions contained in a bidirectional composition can be referred to by the relation records of §7. But this paper does not state a failure state of either ACS from the existence of the composition or the pair of categories as a general consequence. When examining the differences listed in the two descriptors, the holding of blockage and collapse is judged separately per individual trajectory. No rule in this section derives, in general, the failure state of one from that of the other.

Connection to the organizational ACS section (`rem:bidirect_to_sec8`). $\mathrm{Rel}_{ij,t}$ can be material for describing the coupling structure at the organization level treated in §8. But this paper does not state, from a pair of relation descriptors, the establishment of the organization's own self–environment boundary, division and coordination of functions, or an organization-level ACS structure as a general consequence. The section defines neither the judgment that an organizational ACS holds nor the temporal development of networks. Lining up many bidirectional compositions differs from establishing an organization-level ACS structure for the organization itself; the conditions distinguishing an organizational ACS from a mere group of ACSs are given by the later section. Merely enumerating pairs of relations does not show the existence of a new organization-level state space or organization-level feasibility.

Relation quality

Reference: subsec:ro_quality

The subsection formalizes the relation quality $Q_{i\leftarrow j,t}$, which describes in which direction an evaluable relation between Σ_i and an external object $Y_j \in \mathcal{Y}_{-i}$ lies with respect to the maintenance of feasibility of Σ_i, distinguished from the structural environment coupling condition (subsec:env_coupling) and the time-indexed satisfaction predicate of EU (subsec:eu) of §3. As arranged in subsec:env_coupling, this paper treats the structural environment coupling $\mathrm{EC}(x_t)$, the realized external interaction $\chi_{\mathrm{EI},t}$, the time-indexed satisfaction predicate EU_t and the relation quality evaluation Q as mutually different objects. Hence Q introduced here is neither a condition of the structural environment coupling condition nor an additional condition of the time-indexed satisfaction predicate of EU, and Q is not used in the definitions of the time-indexed satisfaction predicates of the three principles. $Q_{i\leftarrow j,t}$ is a classification-C quantity determined under an evaluation rule the describer specifies; the ACS is not required to hold or compute the theoretical value of $Q_{i\leftarrow j,t}$ or of its components in the same numerical representation.

Evaluable domain of relation quality

The external object $Y_j \in \mathcal{Y}_{-i}$ (def:external_object_set) is not limited to other ACSs or agents; non-agent objects, resources, materials, energy sources and other external objects can be included.

Definition (def:Q_rule): evaluation rule of relation quality (classification C). The describer specifies, for each relation (i, j) evaluated, an *evaluation rule* $\mathrm{Rule}_{Q,i\leftarrow j}$. It fixes the interval of operation record or model referred to, the comparison standard, the time width of evaluation, and the criteria for the three components and weights of def:Q, and is fixed over the period of comparison. The rule does not express a normative standard of good and bad, in the same way as the reference standard of a bias (def:bias_tendency). When the time width of evaluation includes records after time t, the value given for t is an after-the-fact description and not an input of the ACS at t (the same treatment as the prediction accuracy coefficient of §5).

Definition (def:Q_domain): evaluable domain of relation quality. The set $\mathcal{D}_{Q,i,t}$ of external objects $Y_j \in \mathcal{Y}_{-i}$ for which $\mathrm{Rule}_{Q,i\leftarrow j}$ gives a value at time t is called the *evaluable domain*. It is a theoretical domain of definition and not the set of objects whose relation quality the ACS can evaluate internally. That Q is theoretically defined is distinguished from the ACS forming and holding an evaluation corresponding to Q internally (subsec:description_levels_global). The direction $i \leftarrow j$ of the index expresses the viewpoint of description (rem:rel_unidirect_notation) and does not mean that Y_j is an agent or that a causal action is realized at time t. Phase 1 does not require $\mathcal{D}_{Q,i,t} \neq \emptyset$ in general.

Not evaluable versus neutral evaluation (rem:Q_undefined). If $Y_j \notin \mathcal{D}_{Q,i,t}$, then $Q_{i\leftarrow j,t}$ is *not defined*. This paper therefore does not assign $Q = 0$ by convention to a relation that cannot be evaluated. The statement $Q_{i\leftarrow j,t} = 0$ is used only when $Y_j \in \mathcal{D}_{Q,i,t}$, the quality evaluation is actually defined, and its overall evaluation is on the neutral boundary. The distinction keeps "not evaluable" different from "neutral quality" explicit.

Formalization of the relation quality evaluation Q

Definition (def:Q): relation quality evaluation Q. At time t let $Y_j \in \mathcal{D}_{Q,i,t}$. Under $\mathrm{Rule}_{Q,i\leftarrow j}$, the overall relation quality that the relation with Y_j has with respect to information formation related to maintaining the feasibility of Σ_i, resource balance and feasibility direction is $Q_{i\leftarrow j,t} \in [-1, 1]$. In Phase 1 it is expressed as $Q_{i\leftarrow j,t} := \alpha^{\mathrm{inf}}_{i\leftarrow j} Q^{\mathrm{inf}}_{i\leftarrow j,t} + \alpha^{\mathrm{res}}_{i\leftarrow j} Q^{\mathrm{res}}_{i\leftarrow j,t} + \alpha^{\mathrm{via}}_{i\leftarrow j} Q^{\mathrm{via}}_{i\leftarrow j,t}$, with each component in $[-1, 1]$. The *information-quality component* $Q^{\mathrm{inf}}_{i\leftarrow j,t}$ expresses, normalized, the positive or negative direction of contribution that informational action or content related to Y_j has to feasibility-fitting information formation in the internal model, evaluation or response formation of Σ_i (> 0: supporting side of information formation relating to feasibility maintenance, < 0: obstructing side). The *resource, material and*

energy balance component $Q^{\mathrm{res}}_{i\leftarrow j,t}$ expresses, normalized, the net direction of contribution of resources, materials, energy and the like on the side of Σ_i accompanying the relation with Y_j to feasibility maintenance (>0: replenishment, acquisition, reduction of load; <0: loss, depletion, increase of load). The *feasibility-direction evaluation component* $Q^{\mathrm{via}}_{i\leftarrow j,t}$ is a formal quantity evaluating the positive or negative contribution of the relation with Y_j to the feasibility margin of Σ_i or to the direction of state change related to feasibility maintenance (>0: supporting direction, <0: impairing direction). The weights satisfy $\alpha^{\mathrm{inf}}_{i\leftarrow j}, \alpha^{\mathrm{res}}_{i\leftarrow j}, \alpha^{\mathrm{via}}_{i\leftarrow j} \geq 0$ and $\alpha^{\mathrm{inf}}_{i\leftarrow j} + \alpha^{\mathrm{res}}_{i\leftarrow j} + \alpha^{\mathrm{via}}_{i\leftarrow j} = 1$. The values of the components and the weights are determined by $\mathrm{Rule}_{Q,i\leftarrow j}$; the weights are formal parameters relative to the object Y_j, its object type and the structure of the ACS, and not universal constants common to all ACSs. For an object for which some of the three components have no meaning as an evaluation axis of the relation, the corresponding weight may be 0, with the sum of weights still 1. Since each component is in $[-1,1]$ and the weights are non-negative with sum 1, Q is a convex combination of the three components and $Q_{i\leftarrow j,t} \in [-1,1]$ holds.

Relation quality versus internal evaluation (`rem:Q_internal_formal`). $Q_{i\leftarrow j,t}$ is a quantity determined by the describer under the evaluation rule and not an internal evaluation of the ACS. The evaluation function for environmental utilization and external coupling of §3 (`def:principle_related_functions`) is a classification-A function identifying and evaluating at least part of the feasibility-related differences accompanying external relations, and the request-by-object evaluation of §4 (`def:rq_eval_types`) is an internal evaluation about internal object tokens. This paper does not require these internal evaluations to agree with $Q_{i\leftarrow j,t}$ (`rem:theory_quantity_vs_internal`). A discrepancy between them (for example $Q_{i\leftarrow j,t} < 0$ for a relation evaluated internally as favorable) does not by itself mean a malfunction.

Relation quality versus properties of the object (`rem:Q_not_property`). $Q_{i\leftarrow j,t}$ is not determined by the properties of Y_j alone but is relative to the ACS, the time and the evaluation rule. For the same Y_j the sign can differ between different ACSs and between different times of the same ACS.

Phase 1 status of the component form of Q (`rem:Q_form`). The three components are a Phase 1 formal representation decomposing relation quality evaluation into three functional viewpoints: information, resource balance and feasibility direction. This does not mean that every biological or artificial ACS holds three independent internal variables, or that the implementation of relation quality evaluation is unique to this three-component decomposition. Operational measurement methods of each component, normalization rules per object type, empirical identification of the weights and higher-dimensional representations of relation quality are not stipulated in Phase 1 (within the present framework).

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.